

BESS Input Cost Assumptions

Senegal PyPSA-Earth model: recalibration of battery storage inputs against real-world benchmarks

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1. Overview

The PyPSA-Earth pipeline sources battery energy storage system (BESS) input costs from the *technology-data* project (v0.13.2), which in turn draws its battery figures from the Danish Energy Agency's (DEA) *Technology Data Catalogue for Energy Storage*. This document records why and how those default assumptions were checked against independent real-world benchmarks, what was found, what was changed, and what genuine software bugs were uncovered and fixed along the way (distinct from the calibration changes).

Two types of BESS representation exist in the model and are both affected by everything below:

- **Existing/committed capacity** — modelled as a `StorageUnit` in `add_electricity.py`, sourced from `powerplants.csv` (Senegal's real, already-installed battery projects).
- **New-build/endogenous capacity** — modelled as a `Store` plus a charger/discharger `Link` pair in `add_extra_components.py`, sized by the optimizer.

All input parameters below are set via `costs`: overrides in each scenario’s config file (`config.uc_bess_{year}.yaml` / `config.uc_no_bess_{year}.yaml`), which flow through `process_cost_data.py`’s cost-processing pipeline into both representations identically (after the architecture fixes described in Section 6).

2. CapEx (investment cost)

2.1 Default (DEA/technology-data)

The DEA catalogue reports two separate cost components:

- battery inverter — the power-conversion hardware, €/kW
- battery storage — the energy-storage medium (cells, racks, enclosure), €/kWh

year inverter (€/kW) storage (€/kWh)

2025	228.63	198.86
2030	170.14	151.00
2035	138.24	125.48
2040	106.34	99.96
2045	85.07	89.86
2050	63.80	79.76

2.2 Benchmark check

Two independent external sources were checked, both expressing cost as a single combined €/kWh figure for a complete 4-hour utility-scale system:

- **NREL**, *Cost Projections for Utility-Scale Battery Storage: 2025 Update* (Cole et al. 2025, NREL/TP-6A40-93281) — a US DOE-funded, bottom-up cost model, Low/Mid/High trajectories, 2024–2060. Explicitly **US-centric**: the underlying cost stack cites Section 301 import tariffs on Chinese-sourced cells/inverters/switchgear, US labour rates, US electricity rates, US sales tax, UFLPA customs-compliance costs, and a partial offset from the 45X US manufacturing tax credit.
- **Ember**, *“How cheap is battery storage?”* (Dec 2025) — a **current-cost snapshot** (not a multi-year forecast) based on real project data: recent auction results in Saudi Arabia, India, and Italy, plus October 2025 industry expert interviews. Reports regional figures for a 4h+ system: China ~\$73/kWh, **rest-of-world (excl. China/US) ~\$125/kWh** (\$75/kWh core equipment + \$50/kWh install & connect), Europe ~\$177/kWh, US ~\$219/kWh.

The DEA-implied combined 4h-equivalent cost was found to run **14–42% above** NREL’s own US-centric mid case (widening with year), and **more than double** Ember’s rest-of-world figure — the most comparable regional basis to Senegal, since it excludes both China’s uniquely subsidised domestic supply chain and the US’s tariff/trade-policy loadings that don’t apply here.

2.3 Methodology: Ember-anchored recalibration

Ember reports a single current-year (2025) figure with no published multi-year trajectory. The correction therefore combines both sources:

1. Take NREL's own Low/Mid/High year-on-year **decline shape**, normalised to 1.0 at 2025.
2. Apply that shape to Ember's lower, regionally-relevant 2025 level (\$125/kWh) instead of NREL's own (higher, US-centric) 2025 starting value.
3. Split the resulting combined €/kWh figure back into separate inverter (€/kW) and storage (€/kWh) prices, using **this model's own DEA-implied power/energy cost ratio** (so the *split logic* — how much of total cost is power hardware vs. energy hardware — is unchanged; only the overall price *level* is corrected) at the same 4-hour duration basis Ember's figure is defined for.

2.4 Final values (applied)

year inverter (€/MW) storage (€/MWh)

2030	88,437	78,488
2035	75,682	68,696
2040	66,962	62,944
2045	55,206	58,311
2050	42,787	53,484

Applied via `costs.investment.{battery inverter, battery storage}` in each scenario config.

Caveat: the config's actual battery duration is `max_hours.battery: 8` (existing fleet; new-build is free to choose). Both NREL and Ember define their headline figures for 4-hour systems, so the blended cost of an 8-hour system built from these same per-unit prices comes out lower per kWh than the reference figures quoted above (the fixed inverter cost is spread over more energy capacity) — this is expected, not an inconsistency.

3. Fixed O&M (FOM)

3.1 Default (DEA/technology-data)

FOM is reported as %/year of investment. Critically, the DEA source only has a FOM figure for battery inverter; **battery storage has no FOM entry in the raw source data at all** — its FOM was effectively zero.

year inverter FOM (%/yr) storage FOM

2025	0.34	(none)
2030	0.34	(none)
2035	0.42	(none)
2040	0.54	(none)
2045	0.68	(none)
2050	0.90	(none)

Note: in **absolute** terms (%/year × the declining CapEx above), this works out to a constant ~€574/MW/year across every year — the rising percentage exactly offsets the declining CapEx base. But since it never applies to the storage side at all — which is 78–83% of total CapEx — **total modelled BESS fixed O&M was understated by roughly 16x** relative to the benchmark below (6% of the correct figure).

3.2 Benchmark check

Two sources converge on **utility-scale BESS annual O&M $\approx$ 2% of CapEx**, but with different scope:

- **Ember**: “Annual operational costs for utility scale battery storage projects are typically low – around 2% of capex... equivalent to \$2.5/kWh/year, which covers routine maintenance, monitoring and performance guarantees, but **excludes augmentation or extended warranties**.”
- **Lazard**’s LCOS assumptions independently fall in the same $\sim$ 2% range.
- **NREL** (same 2025 report as Section 2) uses **4%** instead — deliberately from the high end of the surveyed range, explicitly because “we have adopted a FOM value from the high end and assume that the FOM cost will counteract degradation such that the system will be able to perform at rated capacity throughout its lifetime.” NREL’s 4% is conditional on $\sim$ 1 cycle/day (365 cycles/year): “If the battery is operating at a much higher rate of cycling, then this FOM value might not be sufficient to counteract degradation.”

The 2% vs. 4% gap is not inconsistency between sources — it’s scope. Ember/Lazard = routine maintenance only. NREL = routine maintenance + an implicit degradation/augmentation reserve.

3.3 Methodology

A **fixed** (usage-insensitive) charge is the wrong mechanism for degradation, which is fundamentally throughput-driven — a battery that never cycles doesn’t degrade from cycling. This matters concretely here: this model’s own solved networks show utilisation ranging from **85–154 cycles/year** (uc_bess sector-coupled batch) up to **334 cycles/year** (baseline_bess_2050_overnight, electricity-only) — a $\sim$ 4x spread across scenarios that a single flat FOM figure cannot correctly represent for all of them.

Decomposition adopted: **FOM = Ember’s 2% (routine only)**, with degradation moved to a **usage-scaled VOM** instead (Section 4), which automatically scales with whatever cycling rate the optimizer actually chooses, staying accurate across scenarios rather than being calibrated to one assumed rate.

3.4 Final values (applied)

Flat 2.0%/year, both battery inverter and battery storage, all years. (Not year-specific — it is defined relative to whatever CapEx is in place that year, so a constant percentage is correct by construction.)

4. Variable O&M / degradation cost (VOM)

4.1 Default

No VOM entry exists in the DEA source for either battery inverter or battery storage — genuinely absent, not sourced-and-zero.

4.2 Methodology: cycle-aging cost

Standard formula from the storage-dispatch literature (cf. “*Factoring the Cycle Aging Cost of Batteries Participating in Electricity Markets*”): spread the storage CapEx over the number of cycles the battery is expected to deliver before reaching end-of-life (conventionally 80% state-of-health), charged per MWh discharged:

$\text{degradation_cost (€/MWh)} = \text{storage CapEx (€/MWh, Ember-anchored)} / \text{cycle_life}$

Cycle life: real-world grid-scale LFP (the chemistry underlying all benchmarks used here, confirmed via NREL’s own cost-category breakdown citing “LFP cells from China”) achieves **4,000–6,000 cycles to 80 % SOH** in practice (lab conditions reach higher, 6,000–8,000+; real-world deployment is lower). **5,000 cycles** used as the mid estimate.

4.3 Final values (applied)

year	storage CapEx (Ember-anchored, €/MWh)	degradation cost (€/MWh, ÷5,000 cycles)
2030	78,488	15.70
2035	68,696	13.74
2040	62,944	12.59
2045	58,311	11.66
2050	53,484	10.70

Applied via costs.V0M.battery inverter (see Section 6.3 for why it’s placed there and not on battery storage, and why it’s routed to apply once per MWh *discharged* rather than once per MWh cycled through both charging and discharging).

For context: this is a genuinely material cost, not a rounding term — roughly two-thirds of CCGT’s marginal cost (€45.2/MWh in 2030) or half of OCGT’s (€61.9/MWh).

5. Round-trip efficiency

5.1 Default (DEA/technology-data)

battery inverter,efficiency (labelled “Round trip efficiency DC” in the source): 0.955 (2025) → 0.96 (2030 onward, flat). battery storage has **no efficiency entry at all** — the energy-storage medium was implicitly assumed perfectly lossless.

5.2 Benchmark check

DEA’s 0.96 is specifically the **inverter/PCS component’s own** conversion efficiency, not a whole-system figure — real field data shows installed **whole-system AC round-trip efficiency running 88–94 %** in practice (thermal management/cooling, BMS overhead, and auxiliary loads all cost additional energy that inverter-only efficiency doesn’t capture).

NREL (same report) selects **85 %** explicitly as a whole-system figure: “*Round-trip efficiency is defined as the system efficiency through a charge/discharge cycle... it would include losses associated with cooling systems or battery control equipment... chosen to be 85%, which is well aligned with published values.*”

5.3 On a possible improving trend

Checked specifically whether efficiency should rise over the projection period the way CapEx does. Two considerations:

- Modern PCS/inverter hardware already commonly achieves 96–99% *today* — DEA’s own 2030 figure (96%) is already near that practical range, leaving limited room for the power-electronics conversion step

- itself to improve further (a real physical ceiling effect, unlike CapEx).
- However, the 96%→85% gap is auxiliary/thermal/BMS overhead, not core conversion loss, and that kind of system-integration overhead plausibly has more room to improve with better engineering over 20+ years than raw power-electronics efficiency does.

No sourced year-by-year trajectory was found for this, however — NREL’s own report, the same one providing the detailed year-by-year CapEx trajectory (Table 2), treats round-trip efficiency as a **single flat constant for its entire 2024–2060 projection period**, with no equivalent efficiency-over-time table. Given the standard applied throughout this recalibration (every changed number traces to a specific source, not an assumed curve), the value was kept **flat** rather than applying an unsourced improvement trend.

5.4 Final value (applied)

Flat 0.85, battery inverter only (mechanically reaches both representations via the architecture fix in Section 6.1), all years.

6. Software bugs found and fixed

These are independent of the calibration work above — genuine defects in how the existing code applied *whatever* cost/efficiency values were configured, discovered while investigating the questions above. All are fixed in the codebase (not just documented).

6.1 Existing-battery efficiency silently defaulted to 100% (lossless)

`costs_for_storage()` in `process_cost_data.py` derives a combined battery cost row (used by the existing-fleet `StorageUnit`) from `battery inverter` + `battery storage`, but only ever set `capital_cost/marginal_cost/co2_emissions` on that derived row — never efficiency. The resulting `NaN` was passed straight into `n.madd("StorageUnit", ..., efficiency_store=NaN, efficiency_dispatch=NaN, ...)`, which PyPSA silently filled with its own component-schema default of **1.0** — i.e. every existing battery in every network solved this session was modelled as perfectly, losslessly efficient, with no warning.

Fix: `costs.at["battery", "efficiency"]` is now explicitly set to inherit `battery inverter`’s efficiency.

6.2 New-build battery double-applied round-trip efficiency

`add_extra_components.py` applied the *round-trip* efficiency figure unsplit to **both** the charger and discharger `Link`, compounding it ($0.96 \times 0.96 = 92.16\%$ actual round-trip) instead of reproducing the intended round-trip value.

Fix: each leg now uses `sqrt(round_trip_efficiency)`, matching the convention the existing-battery `StorageUnit` already used (once 6.1 was fixed) — both representations now agree.

6.3 Degradation cost placement (Store vs. Link signed-flow issue)

While wiring in the VOM/degradation cost (Section 4), found that the `Store` component’s own `marginal_cost` applies to its signed power flow `p` (positive discharging, negative charging) — a positive `marginal_cost` there would *reward* charging (negative cost) rather than only charging for discharge.

Fix: the Store’s own `marginal_cost` is explicitly kept at 0. The charger Link’s `marginal_cost` is explicitly kept at 0 too. The degradation VOM is applied **only** on the discharger Link (whose flow is one-directional by construction) — once per MWh actually delivered, matching how the existing-battery `StorageUnit`’s single `marginal_cost` slot already only charges on `p_dispatch` (discharge), not `p_store` (charging), by PyPSA’s own design.

6.4 Existing battery used a uniform assumed duration instead of real per-asset data

`attach_existing_batteries()` applied the single config-wide `max_hours.battery: 8` to **every** existing battery asset, discarding real per-asset `duration/StorageCapacity_MWh` data already present in `powerplants.csv`:

capacity (MW)	real duration (h)	real energy (MWh)
40.0	4.0	160.0
10.0	2.0	20.0
24.0	3.0	72.0
15.0	3.0	45.0

Capacity-weighted real average: **~3.34 hours** — the model had been treating this fleet as uniformly 8-hour, **overstating existing battery energy capacity by ~2.4x** (297 MWh actual vs. 712 MWh as modelled, matching the flat 712 MWh figure observed across every network solved this session).

Fix: `attach_existing_batteries()` now reads each asset’s own duration from `powerplants.csv` (falling back to the config default only where real data is missing/ non-positive).

6.5 Existing battery capital cost was blended at the wrong (uniform) duration

A consequence of 6.4: `costs.at["battery", "capital_cost"]` is a single figure computed by blending inverter + storage costs at the config-wide `max_hours` (8h). Charging every asset this same blended rate regardless of its own real duration overstated the cost of shorter-duration assets (e.g. a real 2h battery billed as if it carried 8h of storage CapEx).

Fix: capital cost is now recomputed per-asset from the un-blended inverter/storage component costs, using each asset’s own real duration:

$$\text{asset_capital_cost} = \text{inverter_capital_cost} + \text{asset_duration} \times \text{storage_capital_cost}$$

Verified against real data (pre-recalibration cost basis, for illustration):

capacity (MW)	duration (h)	old capital_cost (uniform, €/MW/yr)	new capital_cost (per-asset, €/MW/yr)
40.0	4	128,460	76,629
10.0	2	128,460	50,714
24.0	3	128,460	63,672
15.0	3	128,460	63,672

7. Lifetime (checked, not changed)

DEA's inverter lifetime is flat at **10 years** across the entire 2025–2050 projection, while storage lifetime rises (22.5y → 25y → 27.5y → 30y, 2025→2040, then flat). This asymmetry — one component's lifetime improving, the other static — was investigated but not corrected:

- NREL's own whole-system lifetime figure is **15 years**, explicitly described as conservative/low end of the literature (“*existing 4-hour batteries are all new, so no system has yet reached 15 or more years*”), with published values reportedly ranging up toward ~20–25 years.
- NREL's own methodology also keeps lifetime **flat** across their whole projection period — no rising trend, same pattern as the efficiency finding in Section 5.

Not changed, given its smaller impact on total system cost relative to CapEx/FOM/VOM, and because raising the *level* (10y → ~15y) without a sourced *trajectory* would introduce the same kind of unsourced-guess problem avoided elsewhere in this document. Left as a documented open item, not a correction.

8. Summary table — final applied values

year	inverter CapEx (€/MW)	storage CapEx (€/MWh)	FOM (%/yr, both)	VOM/degradation (€/MWh)	round-trip efficiency
2030	88,437	78,488	2.0	15.70	0.85
2035	75,682	68,696	2.0	13.74	0.85
2040	66,962	62,944	2.0	12.59	0.85
2045	55,206	58,311	2.0	11.66	0.85
2050	42,787	53,484	2.0	10.70	0.85

Applied identically across all 10 batch configs (config.uc_bess_{2030,2035,2040,2045,2050}.yaml, config.uc_no_bess_{2030,2035,2040,2045,2050}.yaml) via costs.{investment,FOM,VOM,efficiency} overrides. Lifetime left at DEA's own default trajectory (Section 7).

9. Sources

- Danish Energy Agency, *Technology Data Catalogue for Energy Storage*, as distributed via the PyPSA technology-data project (v0.13.2).
- Cole, W. et al. (2025), *Cost Projections for Utility-Scale Battery Storage: 2025 Update*, National Renewable Energy Laboratory, NREL/TP-6A40-93281. <https://docs.nrel.gov/docs/fy25osti/93281.pdf>
- Ember (Dec 2025), “*How cheap is battery storage?*”. <https://ember-energy.org/latest-insights/how-cheap-is-battery-storage/>
- Lazard, Levelized Cost of Storage (LCOS) analysis — cited within the Ember source above as an independent cross-check on the ~2% O&M figure.
- Factoring the Cycle Aging Cost of Batteries Participating in Electricity Markets (cycle-aging cost methodology reference).

Document prepared from the working session recalibrating BESS input assumptions in the Senegal PyPSA-Earth model. All external figures were independently fetched/verified during that session rather than taken from memory; see the two associated Jupyter notebooks (Input_Cost_EDA.ipynb, Sections 9–12) for the underlying computation and plots.